Star Gates Database: A Summary and Quality Assessment

Report Date: 2026-02-23

Executive Summary

This report provides a comprehensive summary and critical assessment of the Star Gates Database as of its current state. The database contains a total of 34 records intended to document star forts. The analysis reveals a collection of data in a very early and raw stage, characterized by significant incompleteness and structural inconsistencies. Geographically, the existing data points to two distinct clusters of fortifications, one strongly suggested to be in Spain and the other in Croatia, based on coordinate data. However, the absence of explicit location names, historical context, and consistent data entry across most records severely limits the database's current utility for research or analysis. The core of this report focuses on a detailed data quality assessment, highlighting systemic issues such as misplaced data, a high percentage of empty fields, and the implications of relying on unverified, crowdsourced information. The conclusion is that the database represents a foundational but flawed dataset that requires substantial cleaning, validation, and enrichment before it can serve as a reliable resource for the study of star forts.

Database Overview and Composition

The Star Gates Database is a specialized collection intended to catalog star forts, a style of fortification that emerged in the early modern period. In its present form, the database consists of 34 distinct entries. This total number represents the entire corpus of information available for analysis. A preliminary examination immediately reveals that the dataset is in a nascent stage of development. Of the 34 records, a significant portion lacks fundamental identifying information. Only ten of the entries, representing approximately 29% of the database, have been assigned a specific name. The remaining 24 entries are currently anonymous, identified only by their numerical ID within the dataset. This high proportion of unnamed records is the first indicator of the data's overall incompleteness. Furthermore, two records, identified as ID 33 and ID 34, are entirely devoid of any information, appearing as blank entries. These empty records may be placeholders for future data entry or artifacts of the data collection process, but in their current state, they contribute to the total count without adding any substantive value. The initial overview suggests that the database is less a comprehensive catalog and more a collection of raw data points gathered from a single primary source, which is identified as OpenStreetMap.

Geographic Distribution Analysis

A detailed analysis of the geographic data contained within the database, while hampered by formatting issues, reveals the concentration of documented sites in two specific European regions. The geographic information is present for 30 of the 34 records, exclusively in the form of latitude and longitude coordinates. A critical structural error is present across all these records: the longitude value has been incorrectly placed in the “Notes” column, while the “Longitude” column itself remains empty. By manually pairing the “Latitude” and “Notes” columns as coordinate pairs, it becomes possible to map the distribution of these forts.

The first distinct cluster comprises eight records (IDs 1 through 8). The latitude for this group is consistently around 41.6 degrees North, and the longitude is approximately 0.6 degrees East. These coordinates place the cluster in the northeastern region of Spain, in close proximity to the city of Lleida in Catalonia. All eight of these entries are named, with titles such as “Baluard de la Reina,” “Baluard de Louvigny,” and “Punta de Diamant,” which are Catalan names corresponding to known bastions of the main citadel of Lleida. Despite the presence of names and coordinates, the database fails to provide explicit confirmation of the country, region, or city for these entries, as the corresponding fields are uniformly blank.

The second, and larger, cluster consists of 22 records (IDs 11 through 32). These entries are all unnamed. Their coordinates are tightly grouped around 43.5 degrees North latitude and 16.4 degrees East longitude. This geographic location corresponds directly to the city of Split in Croatia, a major port on the Adriatic Sea with a rich history of fortifications, including Venetian-era defenses. The high density of points in such a small area suggests that these records may represent individual components of a larger defensive system, such as separate bastions, walls, or outworks, rather than 22 distinct forts. As with the Spanish cluster, the database lacks any explicit textual information confirming the location as Split or Croatia.

The remaining four records (IDs 9, 10, 33, and 34) contain no coordinate data whatsoever. Two of these have names—“Bedem Cornaro” and “Bedem Priuli,” which are names of bastions in Split, Croatia—but without coordinates, their positions cannot be programmatically mapped or verified against the second cluster. The final two records are completely empty. In summary, the geographic data, once corrected for structural errors, points exclusively to two locations in Spain and Croatia, leaving the global distribution of star forts vastly underrepresented in the current database.

Data Quality and Integrity Assessment

A thorough assessment of the data quality reveals critical deficiencies that currently undermine the database’s reliability and analytical potential. The evaluation can be broken down into several key areas: data completeness, structural accuracy, and the implications of the data’s origin. The most pervasive issue is the profound lack of completeness across nearly all data fields. Of the twelve columns provided, only “ID,” “Latitude,” and “Notes” (erroneously containing longitude) have a significant fill rate. Fields considered essential for historical and geographical context, such as “Country,” “Region,” “City,” “Construction_Date,” “Builder,” “Status,” and “Fort_Type,” are 100% empty for all 34 records. This means that beyond a name for a minority of entries and raw coordinates for most, the database offers no context about who built these fortifications, when they were built, their historical purpose, or their current state of preservation. This absence of information prevents any meaningful historical or architectural analysis, such as timeline distribution or typological comparison.

The second major quality issue is the structural inaccuracy in data formatting. As previously noted, the consistent misplacement of longitude data into the “Notes” field is a fundamental error. This type of systematic mistake suggests a problem in the data extraction or import script used to populate the database. It renders the “Longitude” column useless and requires a manual or programmatic workaround to make the geographic data usable. Any automated process, such as a GIS mapping tool attempting to read the data as-is, would fail. This error calls into question the integrity of the entire data collection and processing pipeline.

Finally, the stated source of the data, OpenStreetMap (OSM), has significant implications for data quality. OSM is a powerful, open-access, and collaborative mapping project that relies on contributions from volunteers. While this can result in incredibly detailed and up-to-date information in well-mapped areas, it also inherently leads to variability in quality, accuracy, and consistency. The data is only as good as the information provided by local contributors. The state of the Star Gates Database reflects this reality. The clusters in Spain and Croatia likely exist because one or more OSM users in those areas took the time to map these features in detail. Conversely, the vast emptiness of the database concerning other regions of the world reflects a lack of data collection from those areas. Furthermore, information from a crowdsourced platform like OSM requires independent verification from authoritative historical and cartographic sources before it can be considered scholarly. The current database appears to be an unverified data dump from the OSM platform, carrying over all the potential inconsistencies and inaccuracies of its source without any additional layer of validation. In its current state, the database is not a reliable analytical tool but rather a preliminary collection of leads that require extensive verification and enrichment.

References

1. OpenStreetMap (https://www.openstreetmap.org/)
2. Overpass API - OpenStreetMap Wiki (https://wiki.openstreetmap.org/wiki/Overpass_API)